

. The atomic radii of the elements are almost same of which series

- (a) Fe-Co-Ni
- (b) Na-K-Rb
- (c) F-Cl-Br
- (d) Li-Be-B

Answer: (a) Fe-Co-Ni

Explanation: The correct choice is Fe-Co-Ni. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

202. In human body if necessary, the plate, screw or wire used for surgery are made up of

- (a) Ni
- (b) Au
- (c) Pt
- (d) Ta

Answer: (d) Ta

Explanation: The correct choice is Ta. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

203. Manganese is related to which block of periodic table

- (a) s-block
- (b) p-block
- (c) d-block
- (d) f-block

Answer: (c) d-block

Explanation: The correct choice is d-block. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

204. A hard and resistant metal (alloy) generally used in tip of nib of fountain pen is

- (a) Os.Ir
- (b) Pt.Cr
- (c) V.Fe
- (d) Fe.Cr

Answer: (b) Pt.Cr

Explanation: The correct choice is Pt.Cr. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

205. Chloride of which of the following elements will be coloured

- (a) Silver
- (b) Mercury
- (c) Zinc
- (d) Cobalt

Answer: (d) Cobalt

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.



206. Which of the following ions has the highest magnetic moment

- (a) Ti^{3+}
- (b) Sc^{3+}
- (c) Mn^{2+}**
- (d) Zn^{2+}

Answer: (c) Mn^{2+}

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; $d0$ and $d10$ species are diamagnetic.

207. Cerium ($Z=58$) is an important member of the lanthanoids. Which of the following statements about cerium is

- (a) The +4 oxidation state of cerium is not known in solutions**
- (b) The +3 oxidation state of cerium is more stable than the +4 oxidation state
- (c) The common oxidation states of cerium are +3 and +4
- (d) Cerium (IV) acts as an oxidizing agent

Answer: (a) The +4 oxidation state of cerium is not known in solutions

Explanation: Across the lanthanoids, 4f electrons shield the nuclear charge poorly, so effective nuclear charge rises and atomic/ionic radii generally decrease.

208. Of the following outer electronic configurations of atoms, the highest oxidation state is achieved by which one of them

- (a) $(n-1)d^3ns^2$
- (b) $(n-1)d^5ns^1$
- (c) $(n-1)d^8ns^2$
- (d) $(n-1)d^5ns^2$**

Answer: (d) $(n-1)d^5ns^2$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before $(n-1)d$ electrons.

209. Among the following series of transition metal ions, the one where all metals ions have $3d^2$ electronic configuration is

- (a) Ti^{4+} , V^{3+} , Cr^{2+} , Mn^{3+}
- (b) Ti^{2+} , V^{3+} , Cr^{4+} , Mn^{5+}**
- (c) Ti^{3+} , V^{2+} , Cr^{3+} , Mn^{4+}
- (d) Ti^{+} , V^{4+} , Cr^{6+} , Mn^{7+}

Answer: (b) Ti^{2+} , V^{3+} , Cr^{4+} , Mn^{5+}

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before $(n-1)d$ electrons.

210. Lanthanoids are

- (a) 14 elements in the sixth period (atomic no. = 58 to 71) that are filling 4f sublevel**



(b) 14 elements in the seventh period (atomic no. = 58 to 71) that are filling 4f sublevel

(c) 14 elements in the sixth period (atomic no. = 90 to 103) that are filling 4f sublevel
14 elements in the seventh period (atomic no. = 90 to 103) that are filling 4f sublevel

Answer: (a) 14 elements in the sixth period (atomic no. = 58 to 71) that are filling 4f sublevel

Explanation: Across the lanthanoids, 4f electrons shield the nuclear charge poorly, so effective nuclear charge rises and atomic/ionic radii generally decrease.

211. Which of the following metals make the most efficient catalyst

- (a) Transition
- (b) Alkali
- (c) Alkaline earth
- (d) Coloured metals

Answer: (a) Transition

Explanation: Transition metals are effective catalysts because they can change oxidation state and form intermediate complexes/adsorb reactants on their surfaces.

212. Lanthanides and actinides resemble

- (a) Electronic configuration
- (b) Oxidation state

- (c) Ionization energy
- (d) Formation of complexes

Answer: (b) Oxidation state

Explanation: Across the lanthanoids, 4f electrons shield the nuclear charge poorly, so effective nuclear charge rises and atomic/ionic radii generally decrease.

213. The lanthanide contraction relates to

- (a) Atomic radii
- (b) Atomic as well as M^{3+} radii
- (c) Valence electrons
- (d) Oxidation states
- (e) Ionisation energy

Answer: (b) Atomic as well as M^{3+} radii

Explanation: Across the lanthanoids, 4f electrons shield the nuclear charge poorly, so effective nuclear charge rises and atomic/ionic radii generally decrease.

214. Which of the following species is expected to show the highest magnetic moment? (At. Nos.: Cr=24, Mn = 25, Co = 27, Ni = 28, Cu = 29)

- (a) Cr^{2+}
- (b) Mn^{2+}
- (c) Cu^{2+}
- (d) Co^{2+}
- (e) Ni^{2+}

Answer: (b) Mn^{2+}



Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

215. Which one belongs to 3d-transition

- (a) Copper
- (b) Gold
- (c) Cobalt
- (d) Silver

Answer: (c) Cobalt

Explanation: The correct choice is Cobalt. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

216. Which one of the following organisation's iron and steel plant was built to use charcoal as a source of power, to start with, but later switched over to hydroelectricity

- (a) The Tata Iron and Steel Company
- (b) The Indian Iron and Steel Company
- (c) Mysore Iron and Steel Limited
- (d) Hindustan Steel Limited

Answer: (a) The Tata Iron and Steel Company

Explanation: The correct choice is The Tata Iron and Steel Company. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

217. Which of the following is the correct sequence of atomic weights of given elements

- (a) Fe>Co>Ni
- (b) Ni>Co>Fe
- (c) Co>Ni>Fe
- (d) Fe>Ni>Co

Answer: (c) Co>Ni>Fe

Explanation: The correct choice is Co>Ni>Fe. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

218. Which of the following element has maximum first ionisation potential

- (a) V
- (b) Ti
- (c) Cr
- (d) Mn

Answer: (d) Mn

Explanation: Compare the stability of the resulting electronic configurations as well as effective nuclear charge; half-filled and fully filled subshells often produce anomalies.

219. A metal M having electronic configuration M -

1s22s22p63s23p63d104s1

- (a) s-block element



(b) d-block element

(c) p-block element

(d) None of these

Answer: (b) d-block element

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

220. Identify the transition element

(a) $1s^2, 2s^2 2p^6, 3s^2, 3p^6, 4s^2$

(b) $1s^2, 2s^2 2p^6, 3s^2, 3p^6 3d^2, 4s^2$

(c) $1s^2, 2s^2 2p^6, 3s^2, 3p^6 3d^{10}, 4s^2 4p^2$

(d) $1s^2, 2s^2 2p^6, 3s^2, 3p^6 3d^{10}, 4s^2 4p^1$

Answer: (b) $1s^2, 2s^2 2p^6, 3s^2, 3p^6 3d^2, 4s^2$

Explanation: The correct choice is $1s^2, 2s^2 2p^6, 3s^2, 3p^6 3d^2, 4s^2$. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

